Smart Attendance System using Face Recognition

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- AI Based Real-Time Attendance System
- Uses Computer Vision & Machine Learning

Introduction

- Traditional attendance is manual & slow
- Time-consuming in large classrooms
- Prone to human errors
- AI enables automation using face recognition

Problem Statement

- Proxy attendance is common
- Manual records are hard to maintain
- Lack of real-time tracking
- Need for secure & automated system

Objectives

- Automate attendance using AI
- Increase accuracy and efficiency
- Eliminate proxy attendance
- Store attendance digitally with timestamp

Methodology

- Capture image using webcam
- Detect faces using OpenCV
- Encode facial features
- Match with stored dataset
- Mark attendance in database

System Architecture

- Input: Camera
- Processing: Face Detection & Recognition
- Matching with database
- Output: Attendance marked in Excel/DB

Technologies Used

- Python Programming Language
- OpenCV for Computer Vision
- NumPy & Pandas for data handling
- Machine Learning Algorithms

Advantages & Limitations

- Advantages:
- Fast, Accurate, Automated system
- No proxy attendance
- Limitations:
- Lighting dependency
- Privacy concerns

Conclusion & Future Scope

- Efficient smart attendance solution
- Reduces manual effort
- Future: Mobile app integration
- Cloud-based storage & deep learning